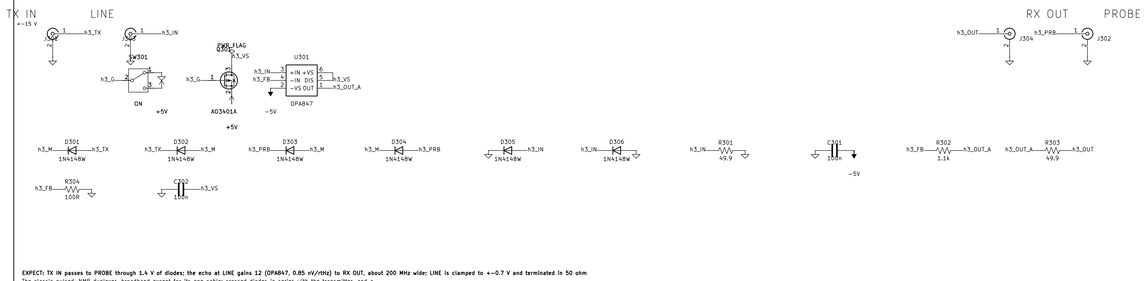
h3 duplexer and preamp, OPA847

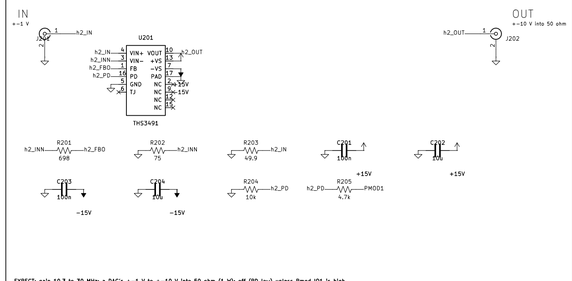
cells (0.0), (1.0)



EXPECT: TX in passes to PROBE through 1.4 V of diodes; the echo at LINE gains 12 (OPA847, 0.85 nV/√Hz) to RX OUT, about 200 MHz wide; LINE is clamped to +−0.7 V and terminated in 50 ohm
The classic pulsed-NMR duplexer, broadband except for its one cable: crossed diodes in series with the transmitter, and a
quarter wave of coax from PROBE to LINE (2.6 m of RG-316 at 20 MHz) whose far end the clamp diodes short during the pulse.
Between pulses the diodes are off; the PA's noise does not reach the probe, the echo reaches the OPA847 (1.1 k / 100 ohm≈
x12. Its minimum stable gain). Without the cable, a short one and a 6 dB pad work at a few dB of loss.

h2 pulse amplifier, THS3491

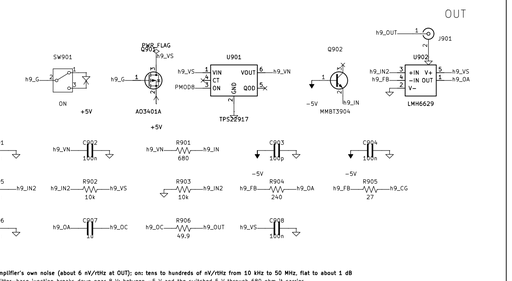
cell (2.0)



EXPECT: gain 10.3 to 30 MHz: a DAC's +−1 V to +−10 V into 50 ohm (1 W); off (PD low) unless Pmod IO1 is high
TI THS3491, current feedback, 900 MHz and 420 mA from +−15 V. 698 ohm from the FB pin (the output, inside) to −IN and
75 ohm to ground set gain 10.3. The 49.9 ohm input termination makes the DAC's +−1 V into 50 ohm the full +−10 V output.
PD from Pmod IO1 (4.7 k, 10 k to ground): high = on. Pulse it: 1 W continuous heats the 3 × 3 mm package past its limit.

h9 noise source, MMBT3904 + LMH6629

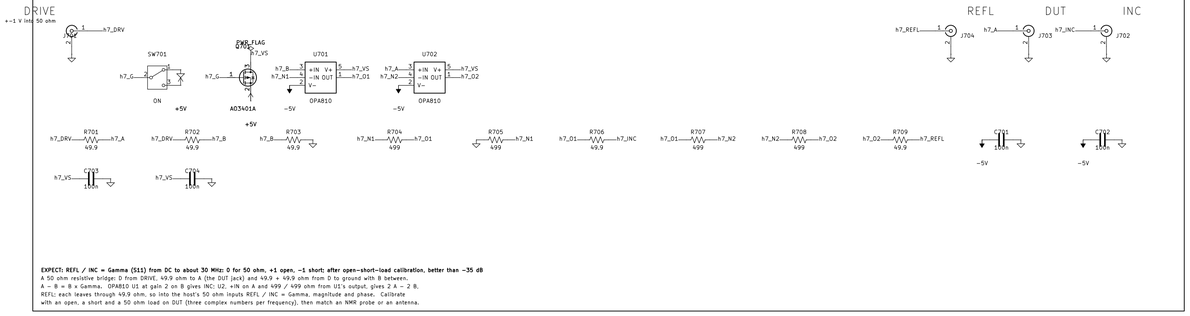
cell (3.0)



EXPECT: off: the amplifier's own noise (about 6 nV/√Hz at OUT); on: tens to hundreds of nV/√Hz from 10 kHz to 50 MHz, flat to about 1 dB
An MMBT3904's emitter-base junction breaks down near 8 V; between −5 V and the switched 5 V through 680 ohm it carries
2–3 mA and its avalanche noise, many times kT/q, goes straight into an LMH6629 at gain 9.9 and out through 49.9 ohm. IOB
high turns it on (a TPS22917 switches its 5 V); hot and cold for a Y-factor noise figure, or a broadband stimulus whose
cross-spectrum against the response on the host's second ADC is a transfer function.

h7 reflection bridge, 2 x OPA810

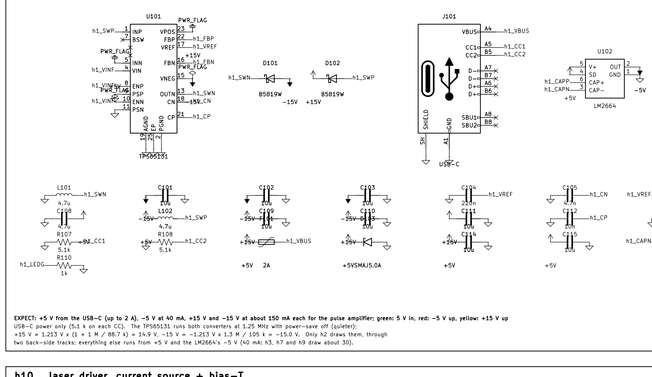
cells (0.1), (1.1)



EXPECT: REFL / INC = Gamma (S11) from DC to about 30 MHz: 0 for 50 ohm, +1 open, −1 short; after open-short-load calibration, better than −35 dB
A 50 ohm resistive bridge: D from DRIVE, 49.9 ohm to A (the DUT jack) and 49.9 + 499 ohm from D to ground with B between.
A − B = B × Gamma. OPA810 U1 at gain 2 on B gives INC; U2, +IN on A and 499 / 499 ohm from U1's output, gives 2 A − 2 B.
REFL: each leaves through 49.9 ohm, so into the host's 50 ohm inputs REFL / INC = Gamma, magnitude and phase. Calibrate
with an open, a short and a 50 ohm load on DUT (three complex numbers per frequency), then match an NMR probe or an antenna.

h1 power, USB-C 5 V in: +5, −5, +15, −15 out

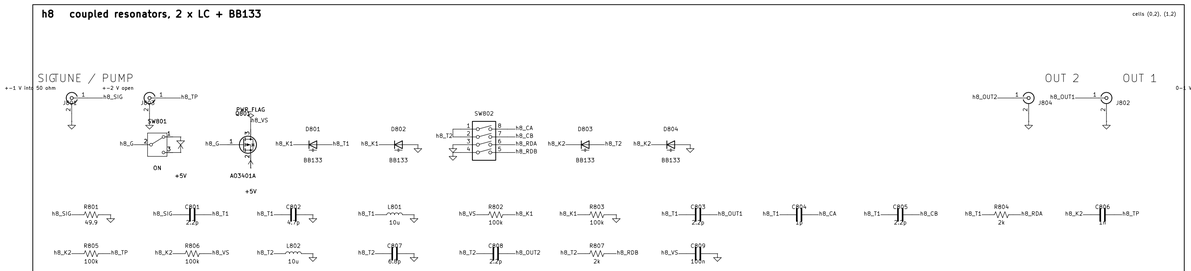
cells (2.1), (3.1)



EXPECT: +5 V from the USB-C (up to 2 A), −5 V at 40 mA, +15 V and −15 V at about 150 mA each for the pulse amplifier; green: 5 V in, red: −5 V up, yellow: +15 V up
USB-C power only (5.1 k on each CC). The TPS65131 runs both converters at 1.25 MHz with power-save off (quieter);
+15 V = 1.213 V × (1 + 1 M / 88.7 k) = 14.9 V, −15 V = −1.213 V × 1.3 M / 105 k = −15.0 V. Only h2 draws them, through
two back-side tracks: everything else runs from +5 V and the LM2664's −5 V (40 mA: h3, h7 and h9 draw about 30).

h8 coupled resonators, 2 x LC + BB133

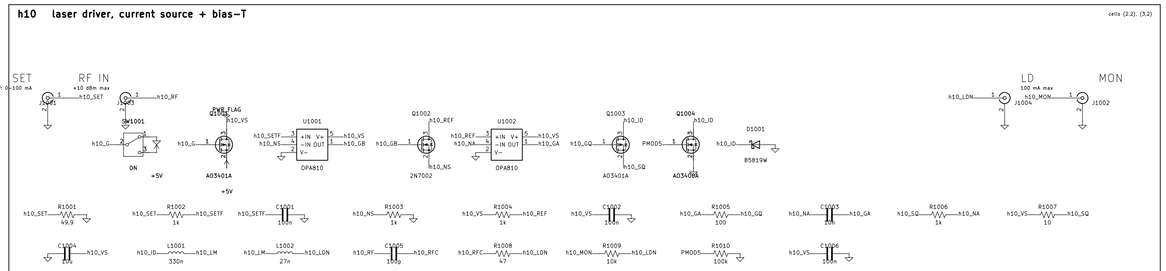
cells (0.2), (1.2)



EXPECT: two 10 MHz tanks (Q about 30); coupled by 1, 2.2 or 3.2 pF they split by 0.4 – 1.2 MHz; TUNE moves tank 2 by +−5 %: a 20 MHz pump on it gives parametric gain
Two LC tanks, 10 uH and about 25 pF each (fixed, two BB133 back to back at 2.5 V, the ports), near 10 MHz. Q about 30.
SIG drives tank 1 through 2.2 pF; each tank feeds its OUT jack through 2.2 pF, −43 dB into the host's 50 ohm. The DIP:
1 pF and 2.2 pF between the tanks (normal modes, the avoided crossing as TUNE sweeps tank 2). 2 k across tank 1 or tank 2
(a lossy tank coupled to a sharp one: the EIT analogue). A pump at twice the frequency on TUNE / PUMP: parametric gain.

h10 laser driver, current source + bias-T

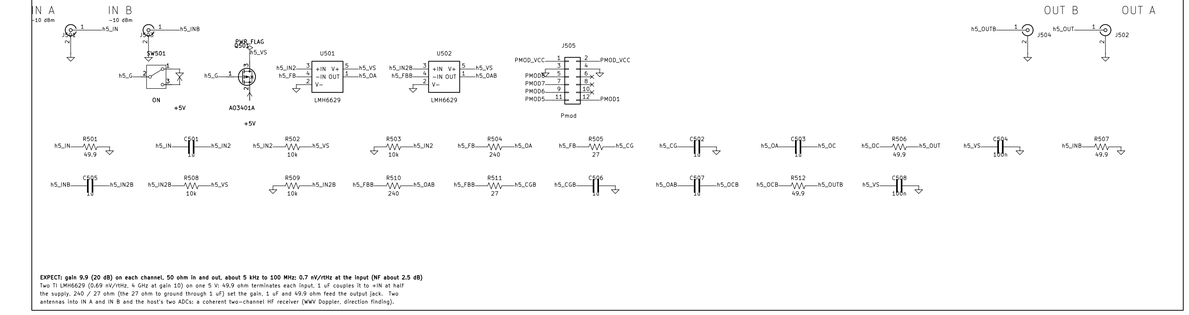
cells (2.2), (3.2)



EXPECT: 0–100 mA from 0–1 V on SET, about 10 kHz of loop bandwidth; IO5 blanks it in about 1 us; RF from about 20 MHz to 3 GHz rides on the current
A high-side current source: REF = VS − SET (OPA810 and a 2N7002 through 1 k / 1 k). OPA810 holds the AO3401's source at
REF so 10 ohm carries SET / 10 ohm. 100 mA per volt (start a VCSEL at a few millivolts). The drain feeds the diode through
330 nH + 27 nH: RF IN joins through 100 pF + 47 ohm; the modulation for a rubidium CPT clock (3.4 GHz) or NV ODMR. IO5 high
shorts the drain (the AO3400) for pulses. MON = the diode's voltage / 200 into 50 ohm: with SET swept, its I–V curve.

h5 HF receiver, 2 x LMH6629

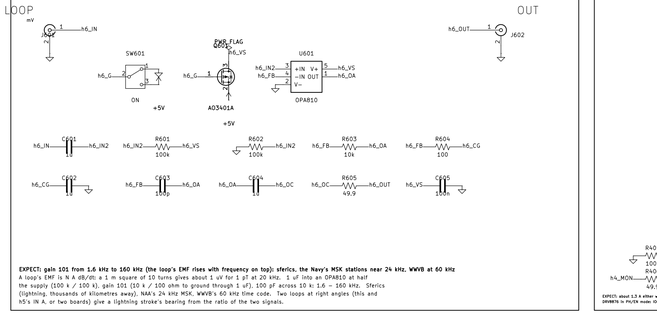
cells (0.3), (1.3)



EXPECT: gain 9.9 (20 dB) on each channel, 50 ohm in and out, about 5 kHz to 100 MHz; 0.7 nV/√Hz at the input (NF about 2.5 dB)
Two TI LMH6629 (0.69 nV/√Hz, 4 GHz at gain 10) on one 5 V: 49.9 ohm terminates each input, 1 uF couples it to +IN at half
the supply, 240 / 27 ohm (the 27 ohm to ground through 1 uF) set the gain. 1 uF and 49.9 ohm feed the output jack. Two
antennas into IN A and IN B and the host's two ADCs: a coherent two-channel HF receiver (WWV Doppler, direction finding).

h6 VLF loop amplifier, OPA810 x101

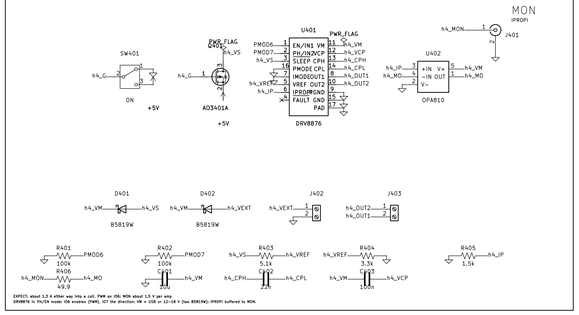
cell (2.3)



EXPECT: gain 101 from 1.6 kHz to 160 kHz (the loop's EMF rises with frequency on top); sferics, the Navy's MSK stations near 24 kHz, WWVB at 60 kHz
A loop's EMF is N A dB/dt: a 1 m square of 10 turns gives about 1 uV for 1 pT at 20 kHz. 1 uF into an OPA810 at half
the supply (100 k / 100 k), gain 101 (10 k / 100 ohm to ground through 1 uF), 100 pF across 10 ki 1.6 – 160 kHz. Sferics
(lightning, thousands of kilometres away), NAA's 24 kHz MSK, WWVB's 60 kHz time code. Two loops at right angles (this and
h5's IN A, or two boards) give a lightning stroke's bearing from the ratio of the two signals.

h4 coil driver, DRV8876

cell (3.3)



EXPECT: about 1.3 A either way into a coil, PWM on IO6; MON about 1.5 V per amp
DRV8876 in PH/EN mode: IO6 enables (PWM), IO7 the direction; OM = USB or 12–18 V (two B5819W); IPROPI buffered to MON.

GND GND
no mounting holes: four stick-on rubber feet, marked on the back

PWR_FLAG
ground loops for the scope's clips, one per channel, in the board's margins

Drawn by scripts/gen_sch.py from boards/phmod.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a pulse amplifier and duplexer, an HF receiver, VLF loops, a bridge, coupled resonators, a coil driver
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/ph
Sheet: /
File: phmod.kicad_sch
Title: PhMOD: magnetic resonance, the ionosphere and light
Size: A1 | Date: 2026-10-25 | **Rev: A**
KiCad E.D.A. 10.0.7 | Id: 1/1